DBA5101 Group Project

Exploring Pricing Strategies for PizzaHub

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Link to code repository

1 Introduction

PizzaHub, a pizza delivery service operating across four regions, introduced targeted pricing experiments on March 1, 2023. The company applied a 5% discount on food prices in Region 3 and a 5% discount on delivery fees in Region 4, while Regions 1 and 2 received no discounts and served as control groups. The objective of this study is to evaluate the causal impact of these two discount strategies on pizza demand and to provide profit-maximizing pricing recommendations for Regions 3 and 4.

Understanding which type of discount—food versus delivery—generates greater demand uplift is critical for PizzaHub's pricing strategy. Additionally, we explore how algorithmic learning methods like multi-armed bandits could enable continuous experimentation and adaptive pricing in the future.

2 Data and Experimental Design

2.1 Dataset Overview

The dataset covers daily order-level data from January 1 to April 30, 2023, with 480 total observations (120 days per region). Key variables include:

- **Date**: Order date
- **Region**: Geographic region (1–4)
- **Unit Price**: Price per pizza ($16 baseline, $15.20 after 5% discount)
- **Quantity Ordered**: Daily pizzas sold
- **Distance**: Delivery distance in miles
- **Delivery Fees**: $0.50 per mile ($0.475 after 5% discount)
- **Discount Applied**: Binary indicator (1 if discount active, 0 otherwise)

Cost structure: Pizza production costs $4 per unit, and delivery costs $0.50 per mile. Fixed costs ($5,000/year) are excluded from our marginal profit analysis.

2.2 Experimental Design

The intervention structure is summarized in Table 1 (see Appendix). This setup allows us to use a difference-in-differences (DiD) approach to isolate the causal effect of each discount type.

2.3 Data Quality Checks

We verified that all 480 observations are complete with no missing values. The discount indicator correctly flags post-March 1 orders in Regions 3 and 4. Unit prices and delivery fees match expected values ($15.20 and $0.475/mile for discounted orders). Preliminary analysis shows that Region 3's average daily demand increased from 9.41 to 15.59 pizzas after the discount, while Region 4's demand rose from 6.64 to 9.92 pizzas per day.

3 Methodology

3.1 Difference-in-Differences Framework

To identify the causal effect of the discounts, we estimate three DiD models:

Model 1 (Food discount effect): Region 3 vs. Regions 1 & 2

$$Q_{it} = \alpha + \delta_{R3} \cdot \text{DID}_{R3,it} + \gamma \cdot \text{post}_t + \varepsilon_{it}$$

Model 2 (Delivery discount effect): Region 4 vs. Regions 1 & 2

$$Q_{it} = \alpha + \delta_{R4} \cdot \text{DID}_{R4,it} + \gamma \cdot \text{post}_t + \varepsilon_{it}$$

Model 3 (Joint model): All four regions

$$Q_{it} = \alpha + \delta_{R3} \cdot \text{DID}_{R3,it} + \delta_{R4} \cdot \text{DID}_{R4,it} + \gamma \cdot \text{post}_t + \varepsilon_{it}$$

where $\text{DID}_{R3,it} = \mathbb{1}(\text{Region}_i = 3) \times \text{post}_t$ and $\text{DID}_{R4,it} = \mathbb{1}(\text{Region}_i = 4) \times \text{post}_t$.
The coefficients δ_{R3} and δ_{R4} measure the average treatment effect on the treated (ATT)—the incremental demand caused by each discount relative to the control regions. Standard errors are clustered by date to account for correlation in daily shocks.

3.2 Parallel Trends Validation

The validity of DiD rests on the parallel trends assumption: in the absence of treatment, treated and control regions would have followed similar demand trajectories. We tested this in two ways:

1. **Pre-period slope test:** We regressed pre-March 1 quantities on time, treatment status, and their interaction. The interaction coefficient (slope difference) was statistically insignificant for both Region 3 ($p = 0.31$) and Region 4 ($p = 0.25$), supporting parallel trends.
2. **Spillover checks:** We tested whether control regions (1 and 2) experienced differential shifts post-March 1. Placebo DiD on controls showed no significant effect ($p = 0.81$), confirming no spillover.

These tests validate our identification strategy (see Appendix for detailed results).

4 Results

4.1 DiD Treatment Effects

Table 2 (see Appendix) presents the main DiD estimates from Model 3 (joint specification). The key results show:

Key findings:

- The 5% food discount in Region 3 increased daily demand by approximately **6.73 pizzas per day** ($p < 0.001$), a statistically significant and economically meaningful effect.
- The 5% delivery fee discount in Region 4 increased demand by **3.82 pizzas per day** ($p < 0.001$), roughly half the magnitude of the food discount.
- The negative *post* coefficient suggests a slight overall decline in control region demand post-March 1, though this is small in magnitude.

These results indicate that **food discounts are approximately twice as effective as delivery discounts** for stimulating demand in this setting.

4.2 Demand Function Estimation

Using the DiD results, we derived linear demand functions for Regions 3 and 4:

Region 3 (price elasticity):

$$Q_3 \approx 144.01 - 8.41 \cdot \text{Price}$$

This implies that a \$1 increase in unit price reduces demand by about 8.41 pizzas per day. The slope estimate comes from dividing the DiD effect (6.73) by the price change ($-\$0.80$).

Region 4 (delivery rate elasticity):

$$Q_4 \approx 83.06 - 152.84 \cdot \text{Rate per km}$$

where the delivery rate per kilometer is derived from the 5% delivery fee discount.

Caveat on optimal pricing: Applying the standard monopoly pricing formula to these linear demand functions yields optimal prices significantly below current levels (e.g., \$10.56 for Region 3). However, this result relies on **linear extrapolation** beyond the observed price range (\$15.20–\$16.00), which may not be reliable. Demand curves often exhibit non-linearities, and consumer behavior at much lower prices is uncertain. Therefore, we do not recommend dramatic price cuts based solely on this calculation. Instead, incremental experiments (e.g., testing 10–15% discounts) would provide better guidance.

4.3 Profit Analysis by Strategy

We computed profit per order as:

$$\text{Profit} = (\text{Unit Price} - 4) \times \text{Quantity} + (\text{Delivery Fees} - 0.5 \times \text{Distance})$$

Table 3 (see Appendix) summarizes performance across the three observed strategies, showing that despite boosting demand, both discount strategies reduce per-order profit compared to no discount (no-discount: \$211.85/order vs. food discount: \$144.26/order vs. delivery discount: \$99.70/order). The food discount maintains higher profitability than the delivery discount, consistent with its stronger demand response.

5 Multi-Armed Bandit for Future Experimentation

While DiD provides retrospective causal estimates, **multi-armed bandit (MAB) algorithms** offer a framework for continuous experimentation. Each pricing strategy is an "arm"; at each order, the algorithm selects a strategy, observes profit, and learns which arm performs best.

We simulated four MAB algorithms (N-Average, ε-Greedy, UCB1, Thompson Sampling) over 2,000 simulated orders, where each round represents one order with profit-per-order rewards sampled from historical data. All algorithms converged to favoring the **no-discount strategy**, which has the highest average profit per order (\$211.85). UCB1 achieved the highest cumulative simulated profit (\$424,032), closely followed by Thompson Sampling (\$423,684) (see Appendix Table 6 and Figure 5).

PizzaHub should implement **UCB1** for ongoing pricing experiments in new regions, as it demonstrated the best performance in our simulation, allowing the company to test new discount levels efficiently and adapt to seasonal or competitive changes in real time.

6 Managerial Recommendations

Based on our analysis, we offer the following recommendations:

6.1 Optimal Pricing Strategy for Profit Maximization

Key Question: What prices maximize profit in Regions 3 and 4?

Using our estimated demand functions and the profit maximization formula $\pi = (P - c) \times Q(P)$, we derive theoretical optimal prices. However, these estimates come with important caveats:

Region 3 (Food Pricing):

- **Theoretical optimal price:** \$10.56 (based on linear demand extrapolation)
- **Problem:** This is 34% below current discounted price (\$15.20) and implies extrapolating demand far beyond observed range
- **Practical recommendation:** Given the unreliability of linear extrapolation, we recommend

incremental testing:
  - Test 10% discount (\$14.40) first, then 15% discount (\$13.60)
  - Monitor profit per day: $\pi = (P - 4) \times Q -$ fixed costs
  - Only proceed to deeper discounts if profit increases
- **Conservative strategy:** Maintain current 5% discount (\$15.20), which balances demand uplift (+6.73 pizzas/day) with acceptable profit margins (\$144/order)

Region 4 (Delivery Pricing):

- **Theoretical optimal delivery rate:** \$0.122/km
- **Problem:** This is below cost (\$0.50/km), resulting in negative delivery margins — economically infeasible
- **Practical recommendation: Eliminate delivery discounts entirely**
  - Delivery discount has weak demand effect (+3.82 pizzas/day) and lowest profitability (\$99.70/order)
  - Revert to full delivery fee (\$0.50/km) to maintain positive margins
  - If demand stimulation is needed, shift budget to food discounts instead

6.2 Summary of Optimal Pricing Strategy

- **Current data favors no-discount pricing overall:** The no-discount strategy yields the highest profit per order (\$211.85 vs. \$144.26 and \$99.70).
- **Region 3:** Maintain or test slightly deeper food discounts (up to 10-15%) with careful profit monitoring. Current 5% discount is a safe choice.
- **Region 4:** Phase out delivery discounts immediately. Use food discounts if demand growth is needed.

6.3 For Demand Growth and Market Penetration

- **Prioritize food discounts:** The 5% food discount nearly doubles demand impact compared to delivery discounts. Use food promotions for new customer acquisition or market entry.
- **Targeted, time-limited promotions:** Instead of permanent discounts, deploy food discounts strategically (e.g., weekdays, new regions, first-time customers) to balance demand growth and profitability.

6.4 Region-Specific Strategy

- **High-delivery-cost regions (like R4):** Delivery discounts are inefficient. Consider transparent pricing (higher delivery fees, lower product prices) or free delivery thresholds instead.
- **Competitive regions:** Use food discounts as a competitive tool, but monitor profit margins closely.

6.5 Continuous Experimentation with MAB

- Implement **UCB1 algorithm** to test new discount levels dynamically in new regions. Our simulation shows UCB1 achieved the highest profit (\$424,032 over 2,000 orders).
- Use MAB to adapt pricing to seasonal demand fluctuations or competitive actions without lengthy A/B test cycles.

7 Conclusion

This study provides strong causal evidence that food discounts are significantly more effective than delivery discounts for increasing pizza demand at PizzaHub. The 5% food discount in Region 3 increased daily demand by 6.73 pizzas, compared to 3.82 pizzas for the delivery discount in Region 4. However, both discounts reduce per-order profitability relative to no-discount pricing.

The key trade-off for PizzaHub is between demand growth and profit margins. If the goal is short-term profit maximization, the data suggest reverting to no-discount pricing or using discounts sparingly. If the goal is market expansion or customer acquisition, food discounts offer better return on investment than delivery discounts.

Looking forward, multi-armed bandit algorithms provide a promising framework for continuous, adaptive experimentation. By implementing MAB in new regions or time periods, PizzaHub can efficiently learn optimal pricing strategies while minimizing opportunity costs.

Limitations: Our demand function estimates assume linearity and are based on a narrow price range ($15.20–$16.00). Extrapolating to much lower or higher prices may be unreliable. Additionally, our 4-month observation period may not capture seasonal effects or long-term customer behavior. Future research should test non-linear demand specifications and longer time horizons.

A Appendix: Tables and Figures

A.1 Tables

Table 1: Intervention Structure by Region

Region	Discount Type	Role	Start Date
1	None	Control	–
2	None	Control	–
3	5% food price	Treatment	March 1, 2023
4	5% delivery fee	Treatment	March 1, 2023

Table 2: DiD Treatment Effects on Daily Pizza Demand

Variable	Coefficient	Std. Error	p-value
DID_R3 (Food discount)	6.730	0.724	0.000***
DID_R4 (Delivery discount)	3.821	0.880	0.000***
post	−0.547	0.254	0.032*
R-squared		0.616	
N		480	

Table 3: Profit Analysis by Pricing Strategy

Strategy	Avg Profit/Order	Total Profit	N Orders
No discount (R1 & R2)	$211.85	$50,844	240
5% food discount (R3)	$144.26	$17,311	120
5% delivery discount (R4)	$99.70	$11,964	120

Variable	Coefficient	Std. Error	z-statistic	p-value
Intercept	17.932	0.185	97.109	0.000***
treat3	−8.525	0.569	−14.980	0.000***
post	−0.547	0.254	−2.152	0.031*
treat3:post	6.730	0.723	9.310	0.000***
R-squared		0.513		
Adj. R-squared		0.509		
N		360		

Standard errors clustered by Date. *p<0.05, **p<0.01, ***p<0.001

Table 4: DiD regression results for Region 3 (food discount) vs. Regions 1 & 2 (controls).

Variable	Coefficient	Std. Error	z-statistic	p-value
Intercept	17.932	0.185	97.109	0.000***
treat4	−11.288	0.685	−16.480	0.000***
post	−0.547	0.254	−2.152	0.031*
treat4:post	3.821	0.879	4.345	0.000***
R-squared		0.646		
Adj. R-squared		0.643		
N		360		

Standard errors clustered by Date. *p<0.05, **p<0.01, ***p<0.001

Table 5: DiD regression results for Region 4 (delivery discount) vs. Regions 1 & 2 (controls).

Policy	Final Cumulative Profit (2,000 orders)
UCB1	$424,032.00
Thompson Sampling	$423,683.63
N-Average	$420,741.00
Epsilon-Greedy	$410,010.48

Table 6: Final cumulative simulated profits for MAB algorithms over 2,000 simulated orders. UCB1 achieved the highest profit, followed closely by Thompson Sampling.

Variable	Mean	Std. Dev.	Min	Max
Quantity Ordered	13.79	6.91	1	35
Unit Price	15.90	0.38	15.20	16.00
Distance	7.00	5.62	2	15
Delivery Fees	3.50	2.81	1.00	7.50

Table 7: Summary statistics for key variables (N = 480).

A.2 Figures

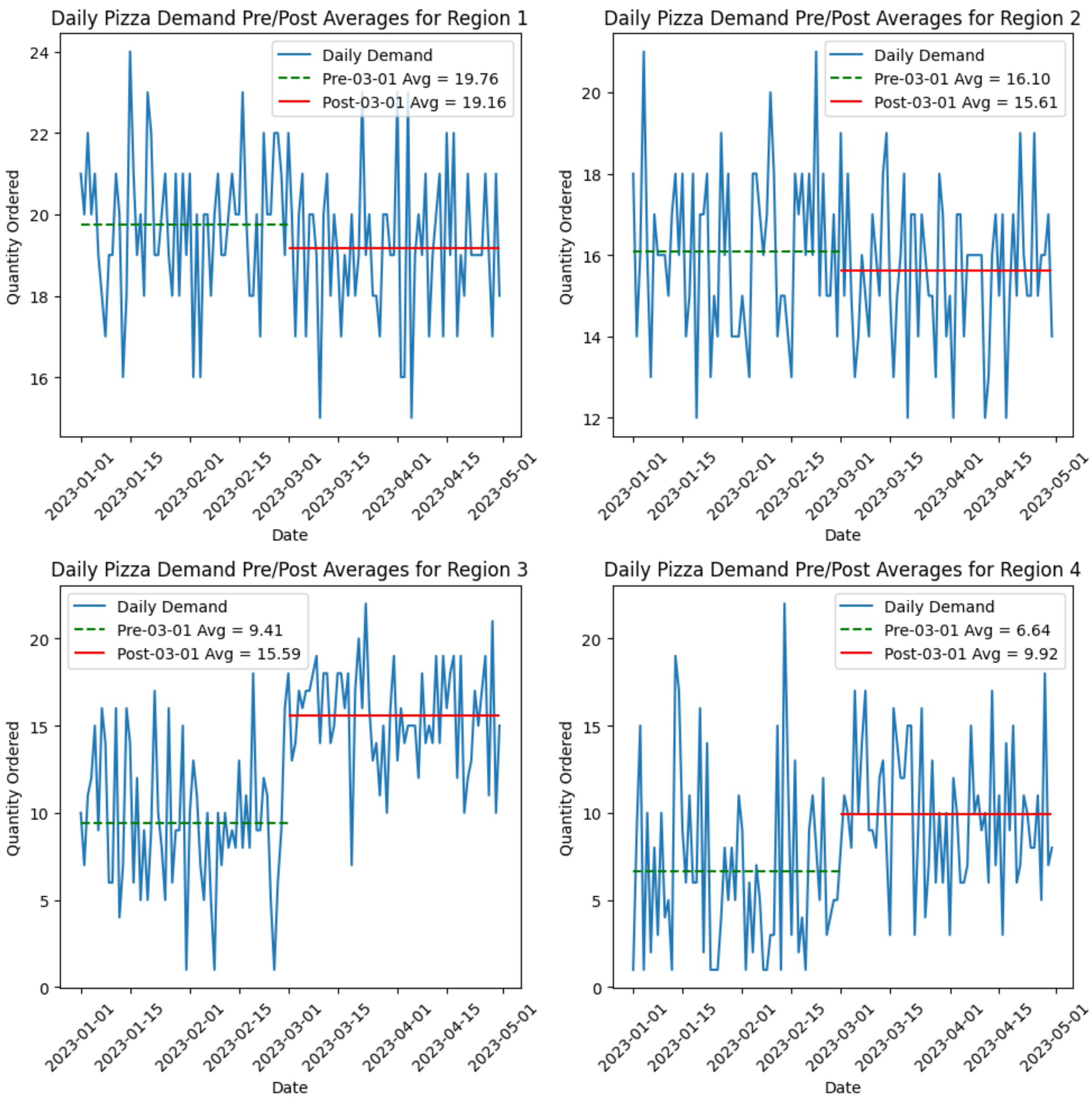

Figure 1: Daily pizza demand with pre- and post-March 1 average lines for each region. Region 3 and 4 show clear demand increases after discount introduction.

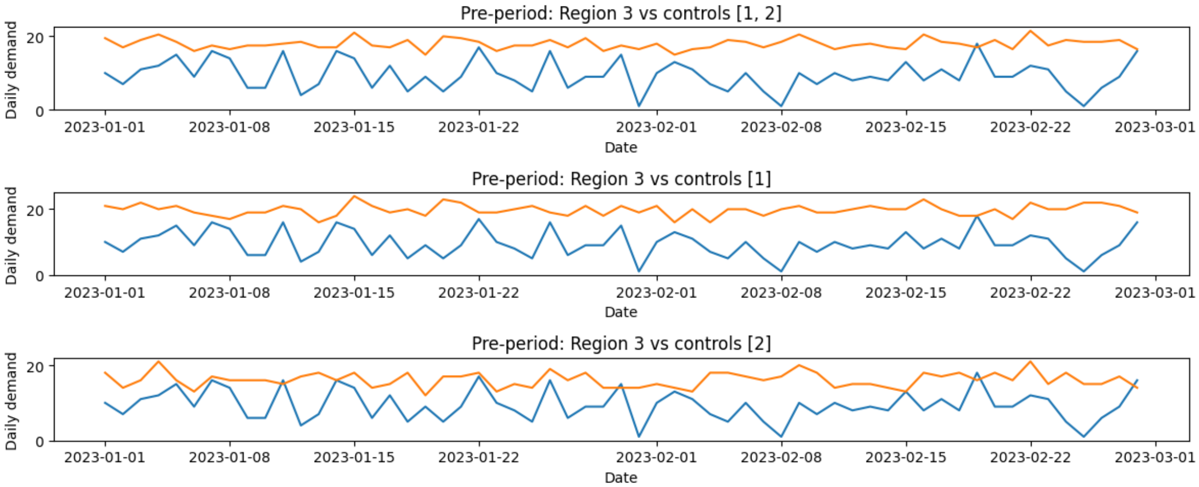

Figure 2: Pre-treatment trends for Region 3 vs. control regions. Slope difference is statistically insignificant (p = 0.31), supporting parallel trends assumption.

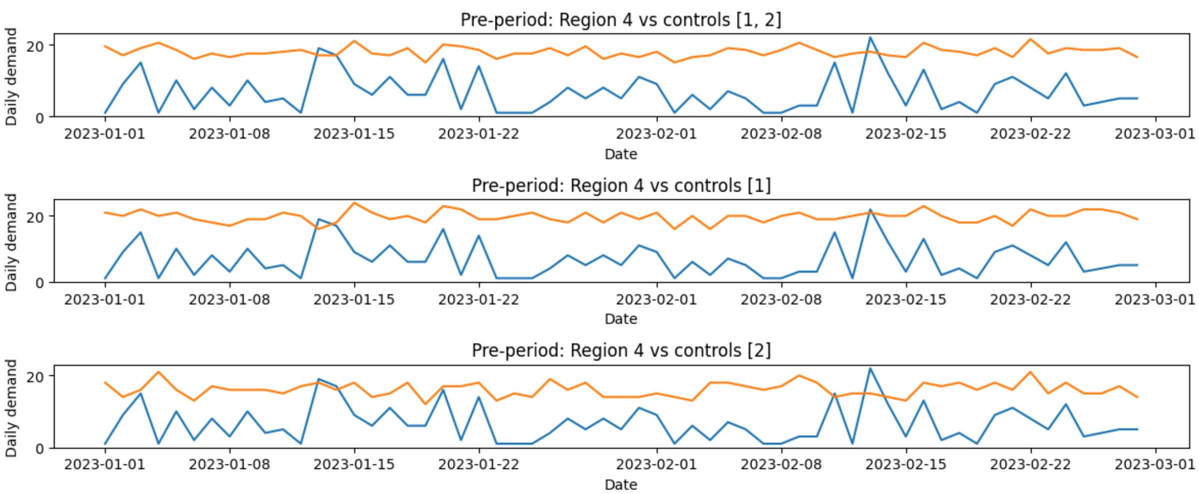

Figure 3: Pre-treatment trends for Region 4 vs. control regions. Slope difference is statistically insignificant (p = 0.25), supporting parallel trends assumption.

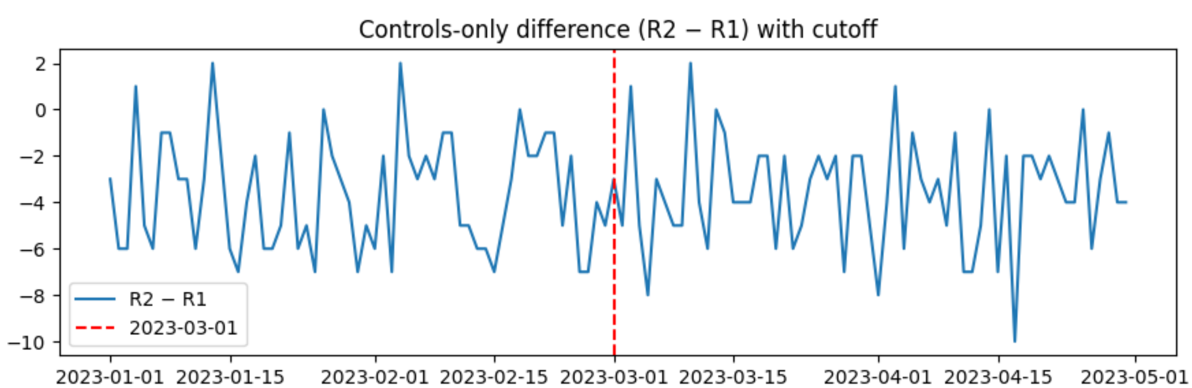

Figure 4: Difference in daily demand between control regions (R2 - R1). No significant breakpoint at March 1 (p = 0.79), indicating no spillover effects.

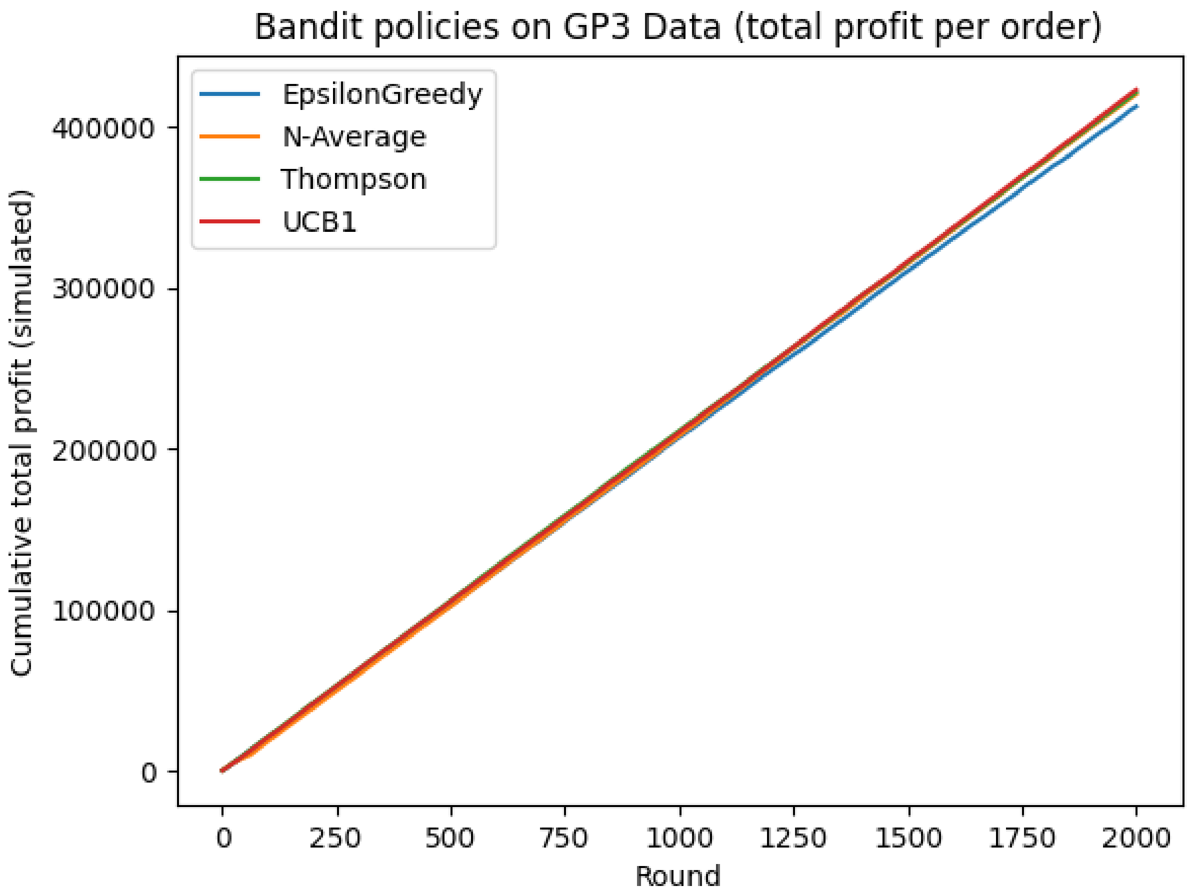

Figure 5: Cumulative simulated profit for four MAB algorithms over 2,000 rounds. UCB1 and Thompson Sampling achieve the highest profits and converge to the no-discount strategy.